Bench exercise: give every crossing an address

Japanese Joinery by Hand · Episode 3. Draw a small board plan (*itazu* 板図), label it with grid addresses (*banzuke* 番付), mark a set of sticks from it, shuffle them, and let the addresses put the frame back together. There's no cutting. About 45 minutes.

1 · Draw the board plan

1. Take a scrap of plywood or stiff card, about A4. This is your board.
2. Rule a grid: 4 lines across and 3 lines down, evenly spaced (say 50 mm, standing in for 910 mm). Draw the grid lines as centre lines.
3. Put a small square at every crossing on the outside of the grid: those are the posts.
4. Draw the sills/beams between neighbouring posts as heavy lines.

2 · Address it

1. Across the top, label the lines い ろ は に (i, ro, ha, ni). Down the side, label them 一 二 三 (1, 2, 3). This is one common layout. Write down which corner you started from: another carpenter might start elsewhere.
2. Write each post's address beside its square: い一, ろ一, … (or i1, ro1, … if you prefer).
3. Name each beam by the two posts it joins, e.g. い一–ろ一.

3 · Mark the "timber"

1. Cut sticks to stand in for posts and beams: chopsticks, dowels or strips of card. Make one per member on your plan.
2. Strike a centre line along each one in pencil.
3. Write each member's address on it, **where it would be hidden once built**: near the bottom of a post, and at each end of a beam, at the joint, on the face the post covers.

4 · Shuffle and rebuild

1. Mix all the sticks in a pile. Hand the pile and the board plan to someone else.
2. Ask them to lay the frame out on the table, flat, from the addresses alone.
3. **Check:** every beam end should sit next to the post with the same address. If any piece was ambiguous, that's the lesson: fix the mark, not the person.

What you should have

- A board plan with a labelled grid, and the corner you started from noted.
- A pile of marked sticks that someone else could assemble without asking you.

The layout, the axes and the starting corner vary by region and carpenter; this sheet uses one common form. See the glossary for the terms.